

Word Combinations

Time limit: 1.00 s

Memory limit: 512 MB

You are given a string of length n and a dictionary containing k words. In how many ways can you create the string using the words?

Input

The first input line has a string containing n characters between a–z.

The second line has an integer k : the number of words in the dictionary.

Finally there are k lines describing the words. Each word is unique and consists of characters a–z.

Output

Print the number of ways modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 5000$
- $1 \leq k \leq 10^5$
- the total length of the words is at most 10^6

Example

Input	Output
ababc 4 ab abab c cb	2

Explanation: The possible ways are ab+ab+c and abab+c.